Gauss' Theorem

Monday, June 9, 2025 6:58 AM

21_ Divergence Theorem of Gauss

21.1

THEOREM 1

Divergence Theorem of Gauss
(Transformation Between Triple and Surface Integrals)

Let T be a closed bounded region in space whose boundary is a piecewise smooth orientable surface S. Let $\mathbf{F}(x, y, z)$ be a vector function that is continuous and has continuous first partial derivatives in some domain containing T. Then

$$\iiint_T \operatorname{div} \mathbf{F}\, dV = \iint_S \mathbf{F} \cdot \mathbf{n}\, dA. \tag{2}$$

***In components** of $\mathbf{F} = [F_1, \; F_2, \; F_3]$ and of the outer unit normal vector $\mathbf{n} = [\cos\alpha, \; \cos\beta, \; \cos\gamma]$ of S* (as in Fig. 253), *formula* (2) *becomes*

$$\begin{aligned}
&\iiint_T \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) dx\, dy\, dz \\
&= \iint_S (F_1 \cos\alpha + F_2 \cos\beta + F_3 \cos\gamma)\, dA \\
&= \iint_S (F_1\, dy\, dz + F_2\, dz\, dx + F_3\, dx\, dy).
\end{aligned} \tag{2*}$$

21.2

Verification of the Divergence Theorem

Evaluate $\iint_S (7x\mathbf{i} - z\mathbf{k}) \cdot \mathbf{n}\, dA$ over the sphere $S: x^2 + y^2 + z^2 = 4$

Soln First we compute the surface integral.

Parameterize S:

$\vec{r} = [2\cos v \cos u, \; 2\cos v \sin u, \; 2\sin v]$, $0 \le u \le 2\pi$, $-\frac{\pi}{2} \le v \le \pi/2$

So $\vec{r}_u = [-2\cos v \sin u, \; 2\cos v \cos u, \; 0]$

$\vec{r}_v = [-2\sin v \cos u, \; -2\sin v \sin u, \; 2\cos v]$

Thus $\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{bmatrix} 4\cos^2 v \cos u \\ 4\cos^2 v \sin u \end{bmatrix}$

Thus $\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{bmatrix} 4\cos^2 v\cos u \\ 4\cos^2 v\sin u \\ 4\cos v\sin v \end{bmatrix}$

and $\vec{F} = \begin{bmatrix} 7x \\ 0 \\ -z \end{bmatrix} = \begin{bmatrix} 14\cos v\cos u \\ 0 \\ -2\sin v \end{bmatrix}$

So then: $\vec{F}\cdot\vec{N} = 56\cos^3 v\cos^2 u - 8\cos v\sin^2 v$

Then $\displaystyle\oiint_S \vec{F}\cdot\hat{n}\,dA = \int_{-\pi/2}^{\pi/2}\int_0^{2\pi} 56\cos^3 v\cos^2 u - 8\cos v\sin^2 v \; du\,dv$

$$= \int_{-\pi/2}^{\pi/2}\int_0^{2\pi} 28\cos^3 v\left(\cos 2u + 1\right) - 8\cos v\sin^2 v \, du\,dv$$

$$= \int_{-\pi/2}^{\pi/2}\left[28\cos^3 v\left(\frac{1}{2}\sin 2u + u\right) - 8u\cos v\sin^2 v\right]_0^{2\pi} dv$$

$$= \int_{-\pi/2}^{\pi/2} 56\pi\cos^3 v - 16\pi\cos v\sin^2 v \, dv$$

$$= \left[56\pi\left(\sin v - \frac{\sin^3 v}{3}\right) - 16\pi\,\frac{\sin^3 v}{3}\right]_{-\pi/2}^{\pi/2}$$

$$= 56\pi\left(1 - \frac{1}{3}\right) - 16\pi\left(\frac{1}{3}\right) + 56\pi\left(1 - \frac{1}{3}\right) - 16\pi\left(\frac{1}{3}\right)$$

$$= 56\pi\left(\frac{2}{3}\right)\times 2 - 16\pi\left(\frac{2}{3}\right) = \pi\left[\frac{112 - 16}{3}\times 2\right]$$

$$= \pi\left[\frac{96}{3}\times 2\right] = \pi\left[32\times 2\right] = 64\pi$$

Next, we use the divergence theorem.

for $\vec{F} = 7x\hat{i} - z\hat{k}$

$$\vec{\nabla}\cdot\vec{F} = \frac{\partial}{\partial x}(7x) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(-z) = 7 + 0 - 1 = 6$$

Now for the sphere $x^2 + y^2 + z^2 \leq 4$:

$$\iiint_V \vec{\nabla}\cdot\vec{F}\,dV = \iiint_V 6\,dV = 6\underbrace{\iiint_V dv}_{\text{Volume of sphere of radius 2}} = 6\left(\frac{4}{3}\pi(2)^3\right) = 6\times\frac{4}{3}\pi\times 8$$

$$= 64\pi$$

sphere of radius 2

21.3 Use the divergence theorem to evaluate $\iint_S \vec{F}\cdot\hat{n}\,dA$ for

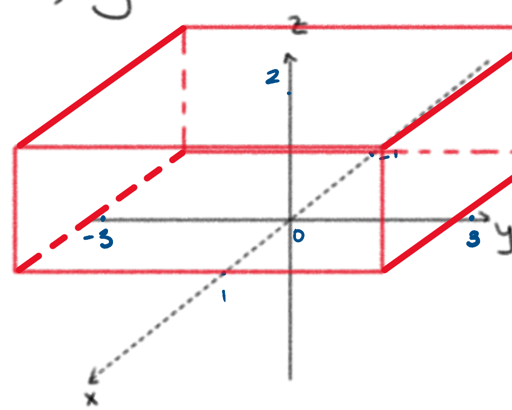
$\vec{F} = [x^2, 0, z^2]$, S: surface of the box $|x| \le 1$, $|y| \le 3$, $0 \le z \le 2$

Soln First we find

$$\vec{\nabla}\cdot\vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z^2) = 2x + 2z$$

The volume is the box, so $-1 \le x \le 1$, $-3 \le y \le 3$, $0 \le z \le 2$

$$\iint_S \vec{F}\cdot\hat{n}\,dA = \iiint_V \vec{\nabla}\cdot\vec{F}\,dV = \int_0^2 \int_{-1}^1 \int_{-3}^3 (2x + 2z)\,dy\,dx\,dz$$



$$= \int_0^2 \int_{-1}^1 \left[2xy + 2zy\right]_{-3}^3 dx\,dz$$

$$= \int_0^2 \int_{-1}^1 (12x + 12z)\,dx\,dz$$

$$= \int_0^2 \left[6x^2 + 12xz\right]_{-1}^1 dz$$

$$= \int_0^2 \left(6(1)^2 + 12(1)z - 6(-1)^2 - 12(-1)z\right) dz$$

$$= \int_0^2 24z\,dz = 12z^2\Big|_0^2 = 12(2)^2 - 0$$

$$= 48.$$